

MARY NJERI MURAGU

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I am a strategic and goal-oriented Python programmer driven by curiosity for data, AI, and automation. I excel in problem-solving, logical analysis, and debugging machine learning projects. I seek to apply my technical and analytical skills to contribute to impactful projects within your organization.

EDUCATION

2023 to present

Bachelor of Computer Science, Jomo Kenyatta University of Agriculture and Technology

Explores a broad range of computer science fields including software development, databases, and networks, with a personal focus on machine learning and AI applications.

EXPERIENCE

April 2025 – September 2025

AI and Machine Learning Backend Developer Intern, JHUB Africa

Developed and trained Python-based models for English-to-Kenyan Sign Language translation and real-time sign capture. Implemented machine learning pipelines for gesture recognition and integrated 3D avatar animations to visualize signed output.

PROJECTS

1. InstruMate Africa | Startup Project

In Progress

Co-founding an EdTech startup building an AI-powered educational toolkit for learners with hearing impairment. Developing English-to-Kenyan Sign Language translation models, real-time gesture capture, and 3D avatar animations for inclusive learning.

2. Resume Matcher API | Personal Project

Built a FastAPI backend that parses resumes and ranks them against job descriptions using NLP techniques and MongoDB. Designed for automated candidate-job matching and integration with HR platforms.

3. Kikuyu–English Translation Model | Research Personal Project

In Progress

Developing a bilingual translation model for Kikuyu and English using neural machine translation. Currently experimenting with Kikuyu speech synthesis for audio output.

4. BugTracker API | Personal Project

Created a RESTful Flask-based bug tracking system using WebSockets for real-time updates and SQLAlchemy for database management. Supports issue reporting, status tracking, and user management.

SKILLS

- Python, FastAPI, Flask
- PostgreSQL, SQL
- HTML, CSS, React
- Three.js, MotionBuilder

Tools

- Git, GitHub, Docker, Linux, Postman, VS code

REFEREES

Referees available upon request.